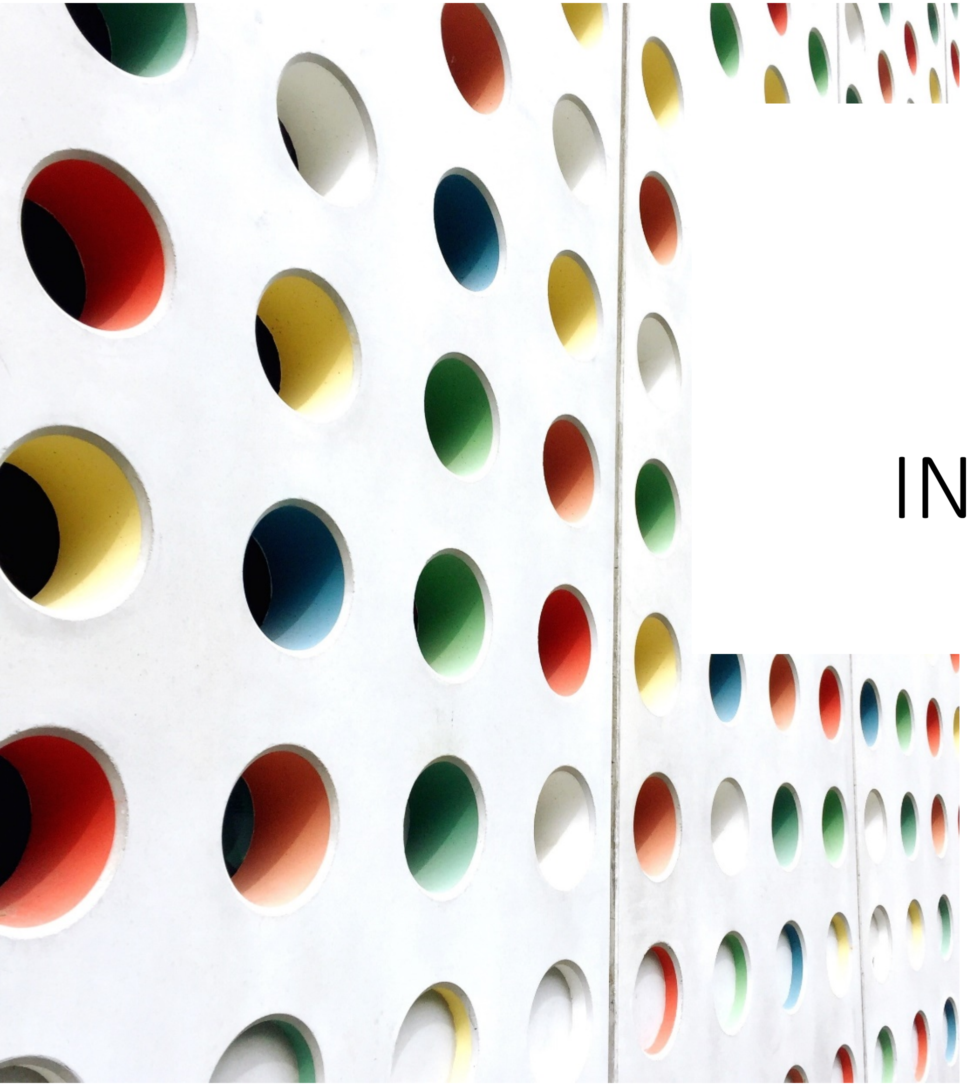
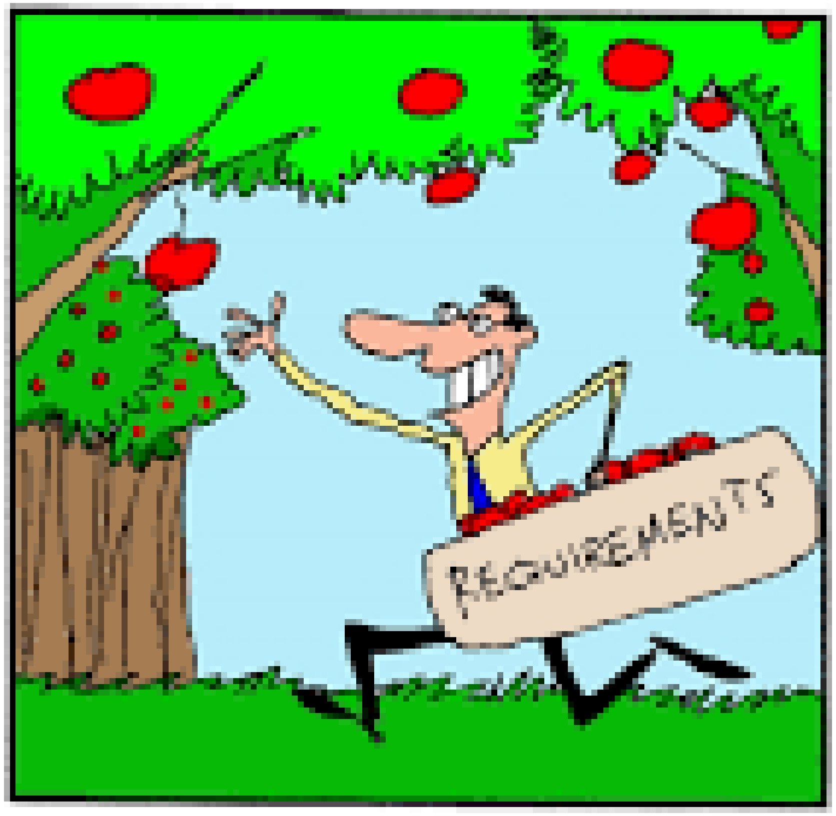


ISP550 INFORMATION SYSTEMS ENGINEERING

Topic 5: **Requirements Elicitation**

OCTOBER 2025 – FEBRUARY 2026

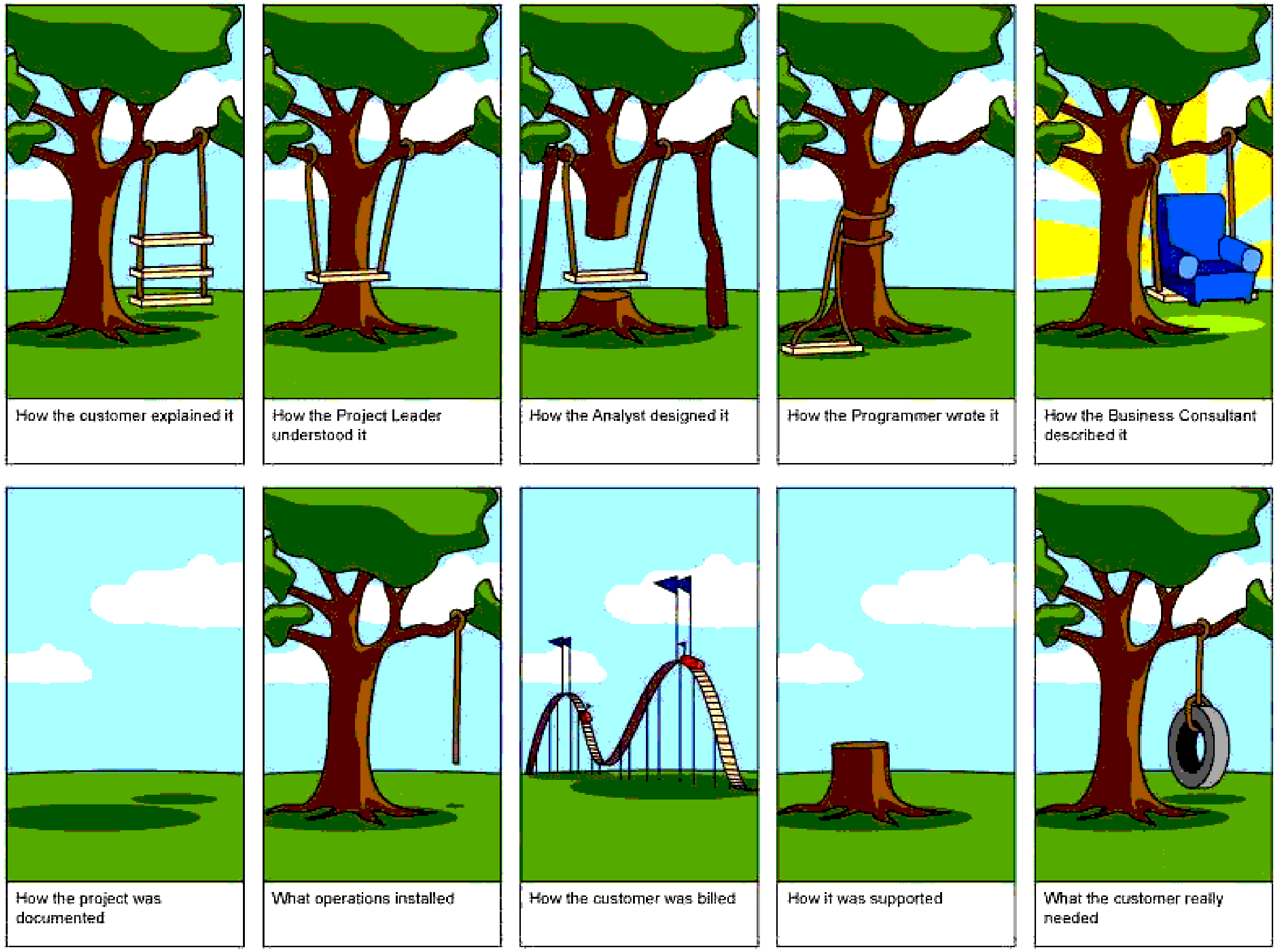
How stakeholders think requirement gathering works.



How requirement gathering really works.

Learning Objectives

- Define Requirements and Requirements Engineering
- Describe Requirements Engineering Activities
- Describe traditional and contemporary requirements elicitation techniques
- Document workflows with activity diagram
- Describe the types of requirements



Requirements are considered as an input to design, implementation and validation phase of software product development. Thus, a software project is successful or a failure during software development because of poor requirement elicitation as well as in requirements managing process (Pfleeger and Atlee, 2006).

In summary, there are many reasons for software project failures; however, poorly engineered requirements process contributes immensely to the reason why software projects fail (Inam, 2015).

Definition of Requirements & Requirements Engineering

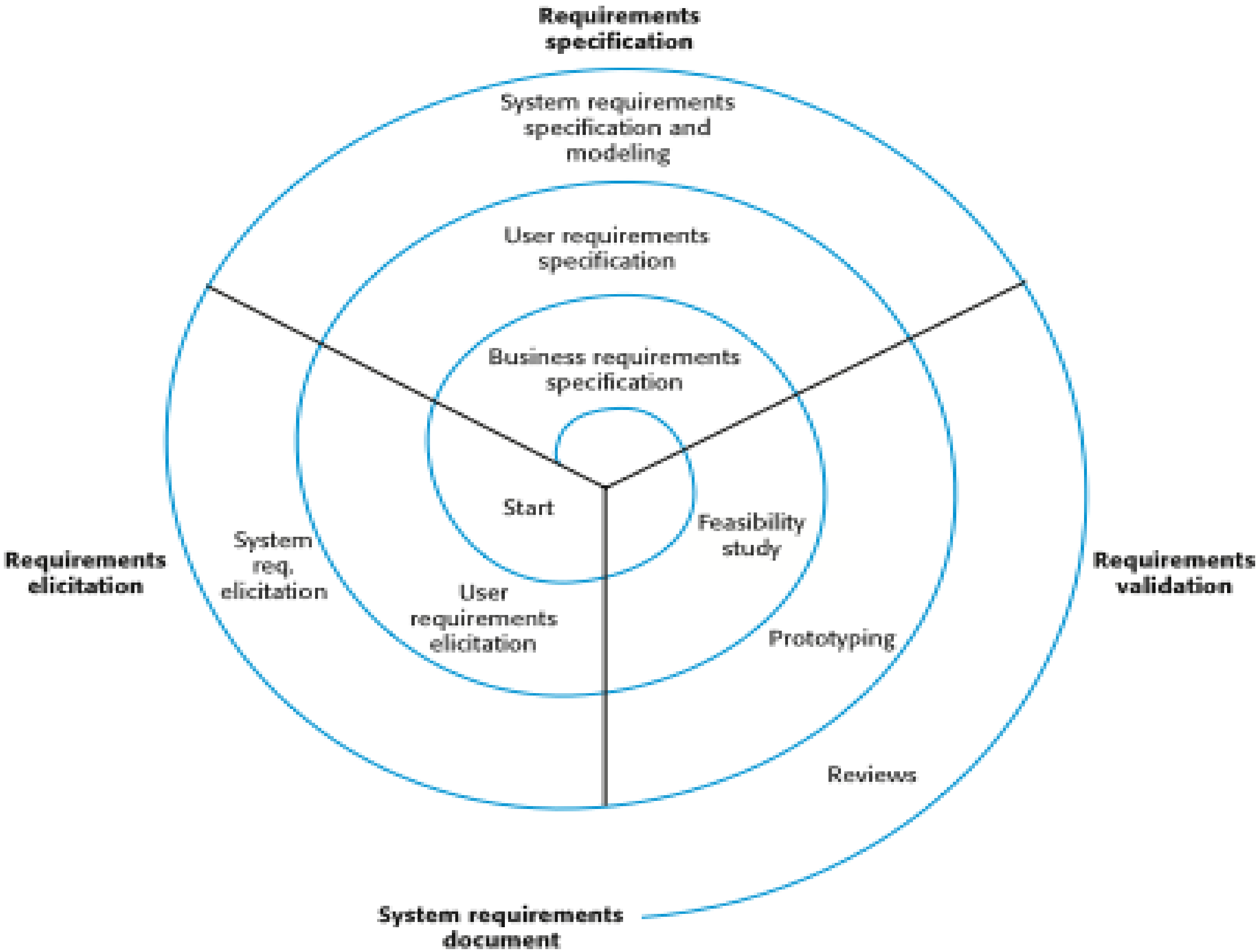
Requirements:
A condition or capability needed by a user to solve a problem or achieve an objective.

Requirements Engineering:
A systematic and disciplined approach to the specification and management of requirements

Requirements Engineering Activities

- The activities used for Requirements Engineering vary widely depending on the application domain, the people involved and the organisation developing the requirements.
- However, there are a number of generic activities common to all processes
  - *Requirements elicitation;*
  - *Requirements analysis;*
  - *Requirements validation;*
  - *Requirements management*.
- In practice, RE is an iterative activity.

A Spiral View of the RE Process



Traditional Methods of Eliciting Requirements

Traditional Methods of Eliciting Requirements
• **Individually interview** people informed about the operation and issues of the current system and future systems needs
• **Interview groups** of people with diverse needs to find synergies and contrasts among system requirements
• **Observe** workers at selected times to see how data are handled and what information people need to do their jobs
• **Study business documents** to discover reported issues, policies, rules, and directions as well as concrete examples of the use of data and information in the organization
• **Distributing questionnaire**

Contemporary Methods of Eliciting Requirements

Contemporary Methods of Eliciting Requirements
Bringing session users, sponsors, analysts, and others together in a **JAD session** to discuss and review system requirements
Iteratively **developing system prototypes** that refine the understanding of system requirements in concrete terms by showing working versions of system features

A JAD is a **Joint Application Development (or Design) session**. It is an opportunity for stakeholders with different points of view to come together to understand business requirements and brainstorm what the best technical approach might be for meeting the customer's needs.

Requirements Elicitation

- **Problems discovered during Requirements Elicitation**
  - **Problems of scope:** The boundary of system is ill-defined. Or unnecessary details are provided.
  - **Problems of understanding:** The users are not sure of what they need, and don’t have full understanding of the problem domain.
  - **Problems of volatility:** the requirements change overtime.

Requirements Elicitation

- **Guidelines of Requirements Elicitation**
  - Assess the business and technical feasibility for the proposed system
  - Identify the people who will help specify requirements.
  - Define the technical environment (e.g. computing architecture, operating system, telecommunication needs) into which the system or product will be placed
  - Identify "domain constraints" (i.e. characteristics of the business environment specific to the application domain) that limit the functionality or performance of the system or product to build
  - Define one or more requirements elicitation methods (e.g. interviews, team meetings, ..etc)
  - Solicit participation from many people so that requirements are defined from different point of views.
  - Create usage scenarios of use cases to help customers/ users better identify key requirements.

Interviewing

- Formal or informal interviews with stakeholders are part of most requirements elicitation processes.
- Types of interview
  - Closed interviews based on pre-determined list of questions
  - Open interviews where various issues are explored with stakeholders.
- Effective interviewing
  - Be open-minded, avoid pre-conceived ideas about the requirements and are willing to listen to stakeholders.
  - Prompt the interviewee to get discussions going using a springboard question, a requirements proposal, or by working together on a prototype system.

Interviewing Users and Other Stakeholders

- Prepare detailed questions
- Meet with individuals or groups of users
- Obtain and discuss answers to the questions
- Document the answers
- Follow up as needed in future meetings or interviews

Themes for Information Gathering Questions

Theme	Questions to users
What are the business operations and processes?	What do you do?
How should those operations be performed?	How do you do it? What steps do you follow? How could they be done differently?
What information is needed to perform those operations?	What information do you use? What inputs do you use? What outputs do you produce?

Preparing for Interview

Checklist for Conducting an Interview

Before

- Establish the objective for the interview.
- Determine correct user(s) to be involved.
- Determine project team members to participate.
- Build a list of questions and issues to be discussed.
- Review related documents and materials.
- Set the time and location.
- Inform all participants of objective, time, and locations.

During

- Arrive on time.
- Look for exception and error conditions.
- Probe for details.
- Take thorough notes.
- Identify and document unanswered items or open questions.

After

- Review notes for accuracy, completeness, and understanding.
- Transfer information to appropriate models and documents.
- Identify areas needing further clarification.
- Thank the participants.
- Follow up on open and unanswered questions.

Interview Session Agenda

Discussion and Interview Agenda

Setting

Objective of Interview
Determine processing rules for sales commission rates

Date,Time, and Location
April 21, 2012, at 9:00 a.m. in William McDougal's office

User Participants (names and titles/positions)
William McDougal, vice president of marketing and sales, and several of his staff

Project Team Participants
Mary Ellen Green and Jim Williams

Interview/Discussion

1. Who is eligible for sales commissions?
2. What is the basis for commissions? What rates are paid?
3. How is commission for returns handled?
4. Are there special incentives? Contests? Programs based on time?
5. Is there a variable scale for commissions? Are there quotas?
6. What are the exceptions?

Follow-Up

Important decisions or answers to questions
See attached write-up on commission policies

Open items not resolved with assignments for solution
See Item numbers 2 and 3 on open items list

Date and time of next meeting or follow-up session
April 28, 2012, at 9:00 a.m.

Interviews in Practice

- Normally a mix of closed and open-ended interviewing.
- Interviews are good for getting an overall understanding of what stakeholders do and how they might interact with the system.
- Interviews are not good for understanding domain requirements
  - Requirements engineers cannot understand specific domain terminology;
  - Some domain knowledge is so familiar that people find it hard to articulate or think that it isn’t worth articulating.

Distribute and Collect Questionnaires

RMO Questionnaire

This questionnaire is being sent to all telephone-order sales personnel. As you know, RMO is developing a new customer support system for order taking and customer service.

The purpose of this questionnaire is to obtain preliminary information to assist in defining the requirements for the new system. Follow-up discussions will be held to permit everybody to elaborate on the system requirements.

Part I. Answer these questions based on a typical four-hour shift.

1. How many phone calls do you receive?________________________________
2. How many phone calls are necessary to place an order for a product?________________
3. How many phone calls are for information about RMO products, that is, questions only?__________
4. Estimate how many times during a shift customers request items that are out of stock.__________
5. Of those out-of-stock requests, what percentage of the time does the customer desire to put the item on back order?______________%
6. How many times does a customer try to order from an expired catalog?________________
7. How many times does a customer cancel an order in the middle of the conversation?__________
8. How many times does an order get denied due to bad credit?____________________

Part II. Circle the appropriate number on the scale from 1 to 7 based on how strongly you agree or disagree with the statement.

Question	Strongly Agree						Strongly Disagree
It would help me do my job better to have longer descriptions of products available while talking to a customer.	1	2	3	4	5	6	7
It would help me do my job better if I had the past purchase history of the customer available.	1	2	3	4	5	6	7
I could provide better service to the customer if I had information about accessories that were appropriate for the items ordered.	1	2	3	4	5	6	7
The computer response time is slow and causes difficulties in responding to customer requests.	1	2	3	4	5	6	7

Part III. Please enter your opinions and comments.

Please briefly identify the problems with the current system that you would like to see resolved in a new system.

____

____

____

Review Inputs, Outputs, and Procedures

Ridgeline Mountain Outfitters—Customer Order Form

RIDGELINE MOUNTAIN OUTFITTERS

Name and address of person placing order.
(Please verify your mailing address and make correction below.)
Order Date __/__/__

Name

Address Apt. No

City State Zip

Phone: Day ( )______ Evening ( )______

Gift Order or Ship To: (Use only if different from address at left.)

Name

Address Apt. No

City State Zip

Gift ☐ Address for this Shipment Only ☐ Permanent Change of Address ☐

Gift Card Message

Delivery Phone ( )______

	Item No.	Description	Style	Color	Size	Sleeve Length	Qty	Monogram	Style	Price Each	Total

MERCHANDISE TOTAL ______

Regular FedEx shipping $4.50 per U.S. delivery address ______
(Items are sent within 24 hours for delivery in 2 to 4 days)

Please add $4.50 per each additional U.S. delivery address ______

FedEx Standard Overnight Service ______

Any additional freight charges ______

International Shipping (see shipping information on back) ______

Method of Payment

Check/Money Order ☐ Gift Certificate(s) ☐ AMOUNT ENCLOSED $ ______

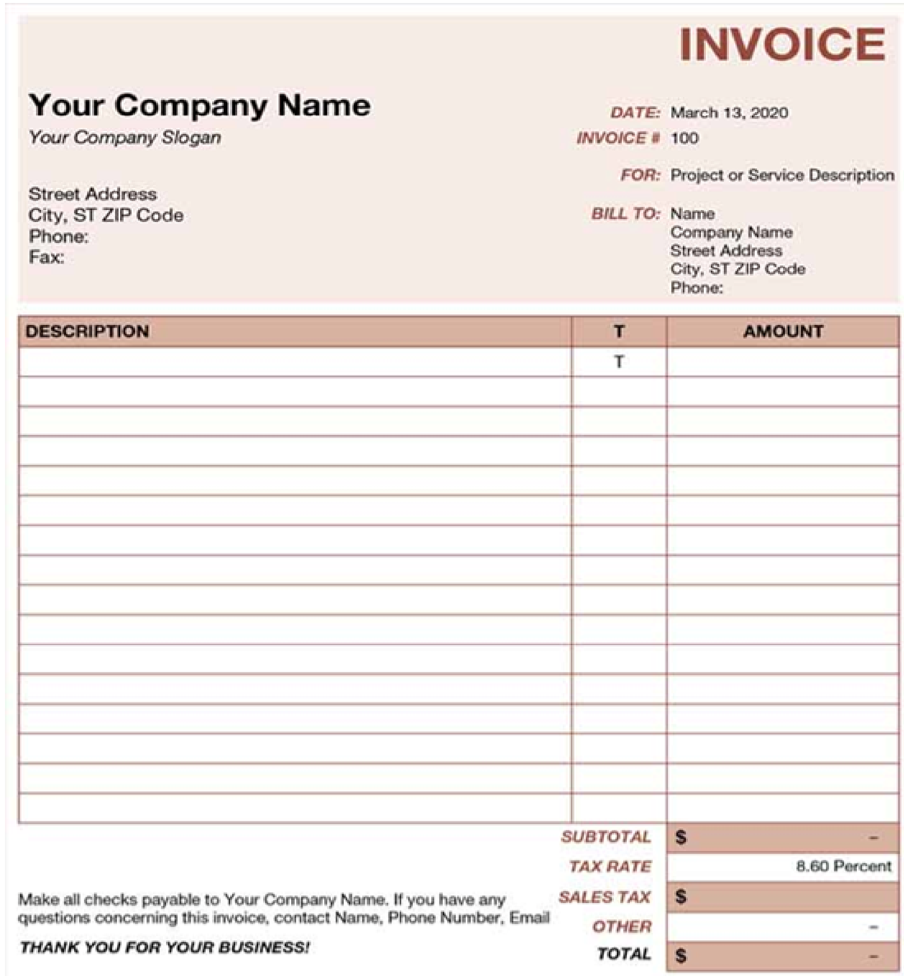
American Express ☐ MasterCard ☐ VISA ☐ Other ☐

Account Number

MO YR
__/__
Expiration Date

Signature

An Invoice Form from Microsoft Excel



INVOICE

Your Company Name
Your Company Slogan

Street Address
City, ST ZIP Code
Phone:
Fax:

DATE: March 13, 2020
INVOICE # 100

FOR: Project or Service Description

BILL TO: Name
Company Name
Street Address
City, ST ZIP Code
Phone:

DESCRIPTION	T	AMOUNT
	T	

SUBTOTAL	$ -
TAX RATE	8.60 Percent
SALES TAX	$
OTHER	-
TOTAL	$ -

Make all checks payable to Your Company Name. If you have any questions concerning this invoice, contact Name, Phone Number, Email

THANK YOU FOR YOUR BUSINESS!

(**Source**: Microsoft Corporation)

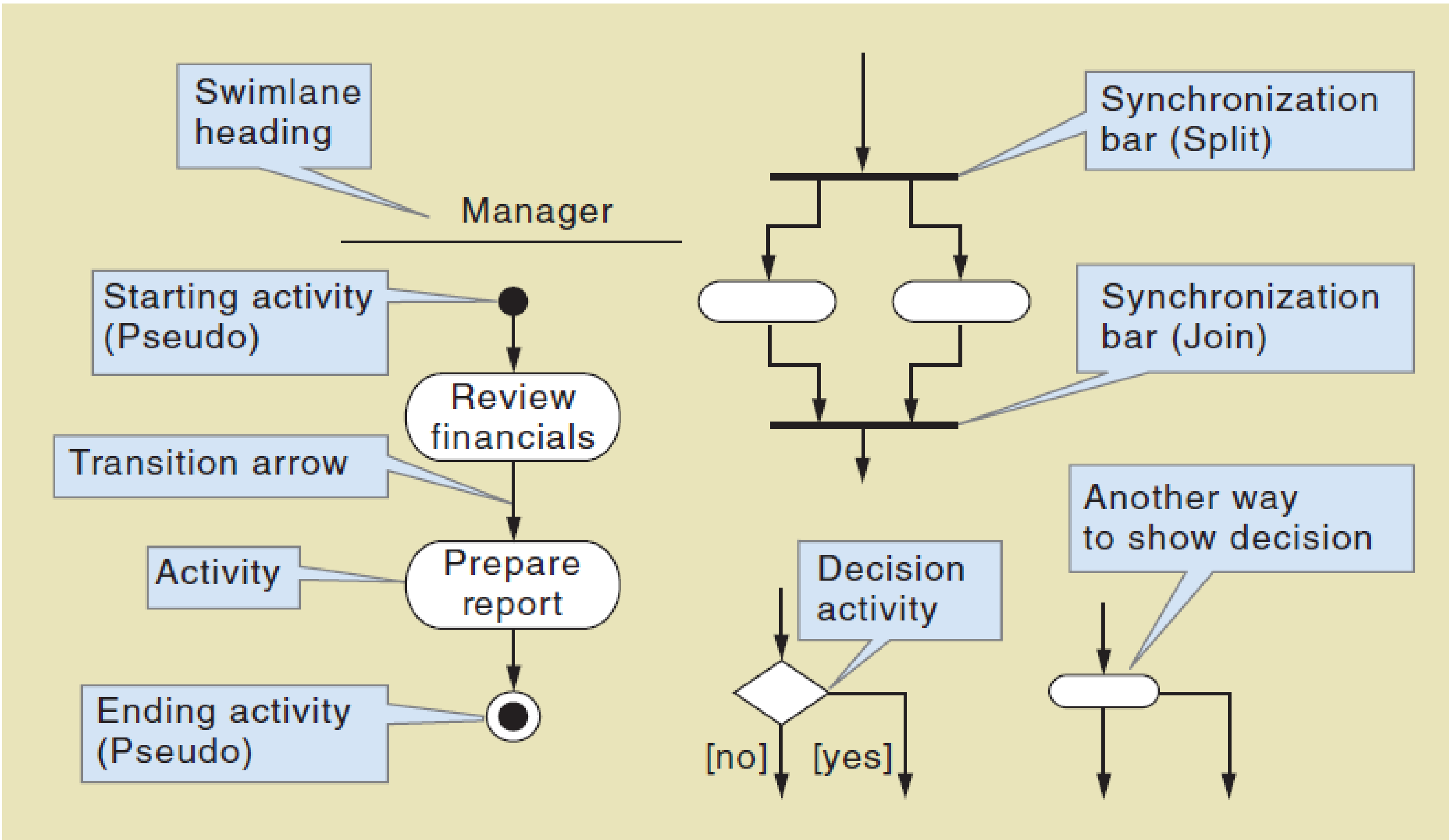
Comparison of Observation and Document Analysis

Characteristic	**Observation**	**Document Analysis**
Information Richness	High (many channels)	Low (passive) and old
Time Required	Can be extensive	Low to moderate
Expense	Can be high	Low to moderate
Chance for Follow-Up and Probing	Good: probing and clarification questions can be asked during or after observation	Limited: probing possible only if original author is available
Confidentiality	Observee is known to interviewer; observee may change behavior when observed	Depends on nature of document; does not change simply by being read
Involvement of Subject	Interviewees may or may not be involved and committed depending on whether they know if they are being observed	None, no clear commitment
Potential Audience	Limited numbers and limited time (snapshot) of each	Potentially biased by which documents were kept or because document was not created for this purpose

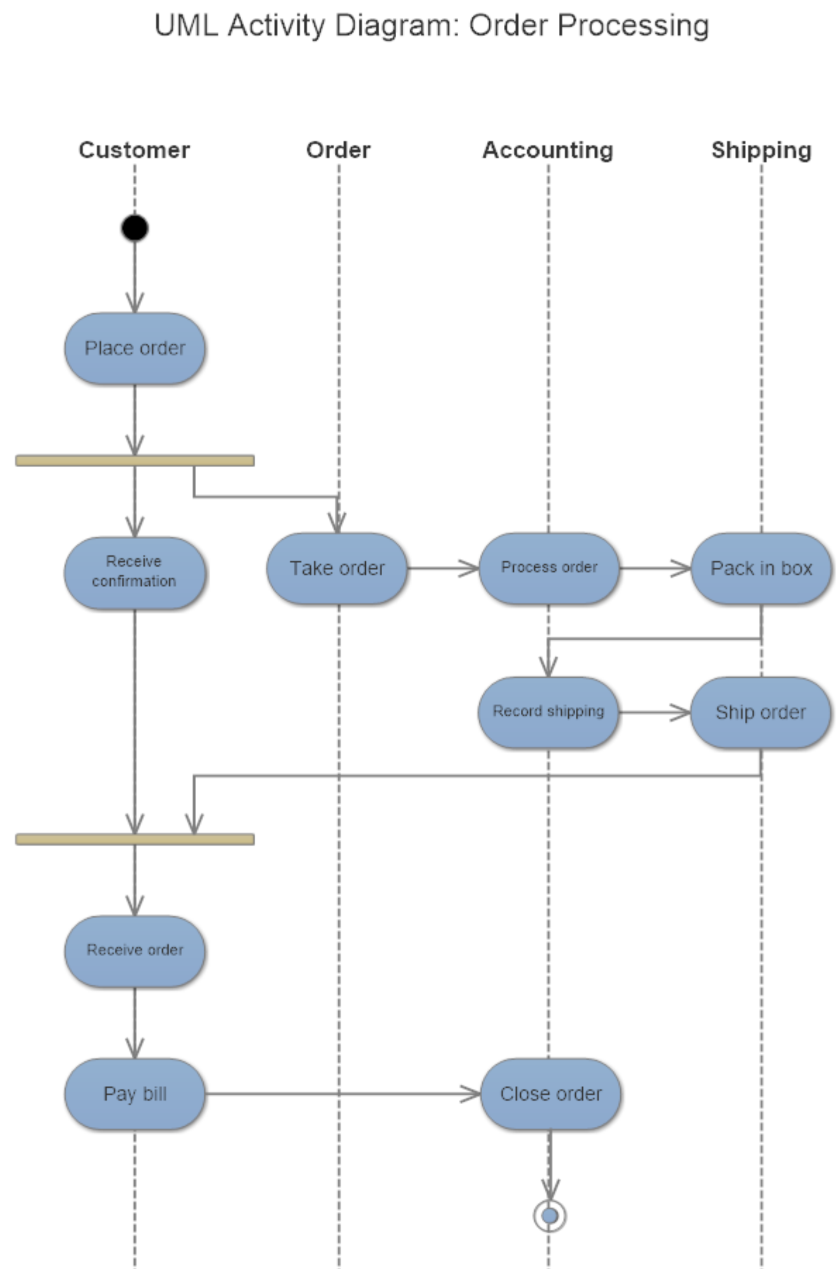
Documenting Workflows with Activity Diagrams

- Workflow– sequence of processing steps that completely handles one business transaction or customer request
- Activity Diagram– describes user (or system) activities, the person who does each activity, and the sequential flow of these activities
  - Useful for showing a graphical model of a workflow
  - A UML diagram

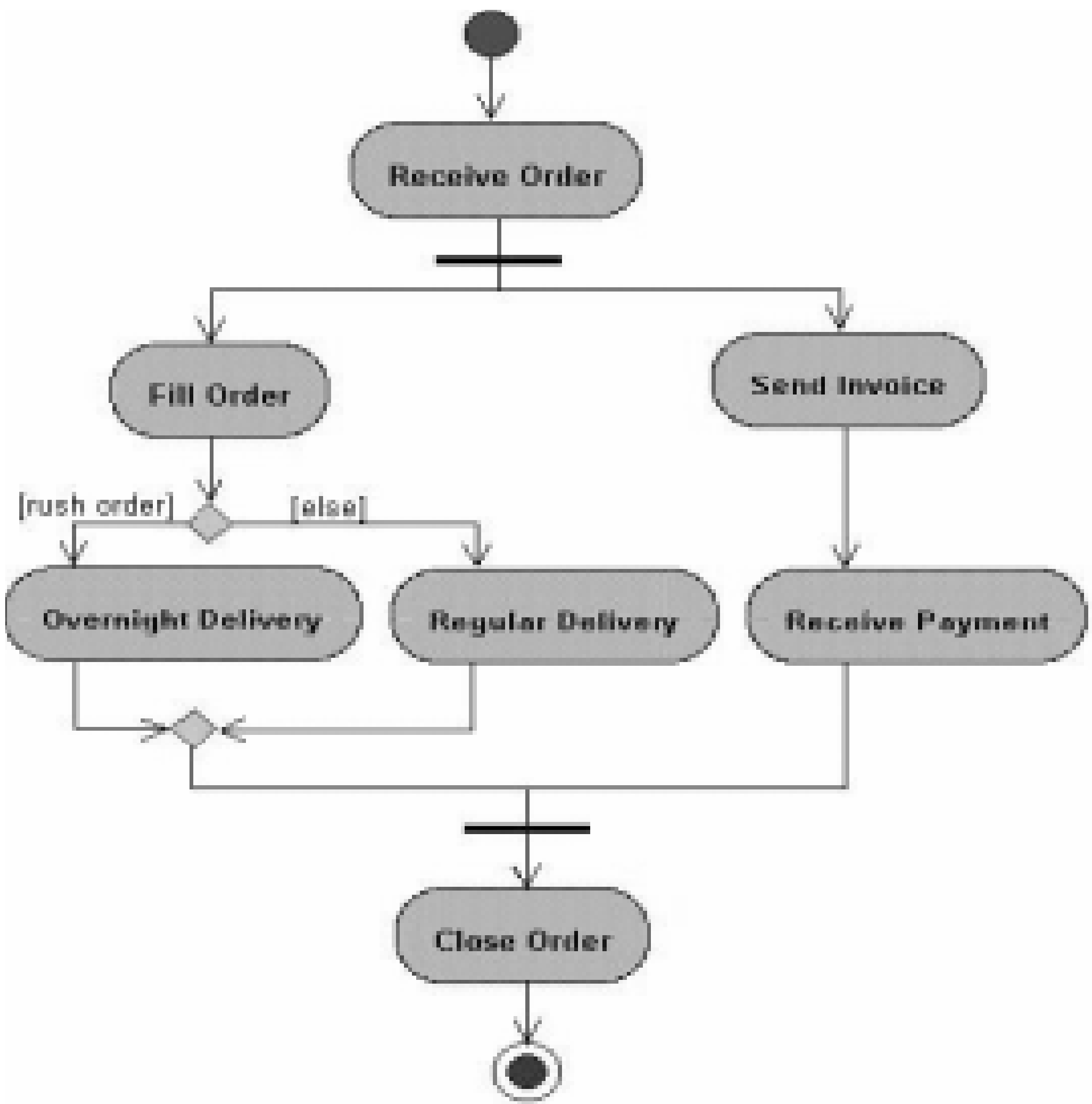
Activity Diagrams Symbols



Activity diagram example (1



Activity diagram example (2)



Types of Requirements (1/2)

- **Functional Requirements**:
  - Defines functions of a software system or its components. They may be calculations, technical details, data manipulation and processing and other specific functionality that define "what a system is supposed to accomplish?"
  - They describe particular results of a system.
  - Functional requirements are supported by Non-functional requirements.

Types of Requirements (2/2)

- Non-Functional Requirements:
  - They are requirements that specify criteria that can be used to judge the operation of a system, rather than specific behavior.
  - Functional requirements define what the system is supposed to ***do***, whereas non-functional requirements define how a system is supposed to ***be***.
  - Non-functional requirements can be divided into two main categories:
    - Execution qualities, such as security and usability, which are observable at runtime.
    - Evolution qualities, such as testability, maintainability and scalability.

FURPS+ Requirements Acronym

- Functional requirements
- Usability requirements
- Reliability requirements
- Performance requirements
- Security requirements
- + even more categories…

FURPS+ Requirements Acronym

Requirement categories	FURPS + categories	Example requirements
Functional	Functions	Business rules and processes
Nonfunctional	Usability	User interface, ease of use
	Reliability	Failure rate, recovery methods
	Performance	Response time, throughput
	Security	Access controls, encryption
	+ Design constraints	Hardware and support software
	Implementation	Development tools, protocols
	Interface	Data interchange formats
	Physical	Size, weight, power consumption
	Support	Installation and updates

- ✓Define Requirements and Requirements Engineering
- ✓Describe Requirements Engineering Activities
- ✓Describe traditional and contemporary requirements elicitation techniques
- ✓Document workflows with activity diagram
- ✓Describe the types of requirements